

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

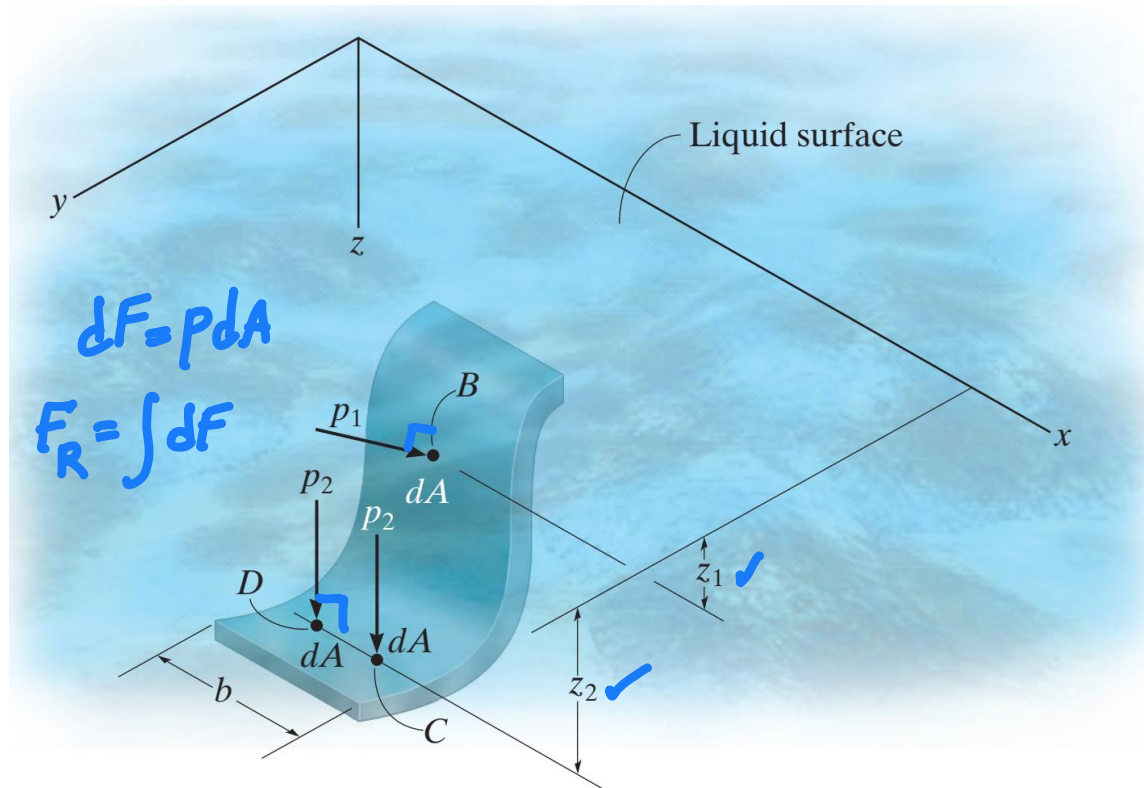
Lecture 16:

Fluid Statics

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Fluid Pressure

According to **Pascal's** law, a fluid at rest creates a pressure p at a point that is the *same* in *all* directions. Also, the magnitude of p , measured as a force per unit area, depends on the specific weight γ or mass density ρ of the fluid, and the depth z of the point from the fluid surface.



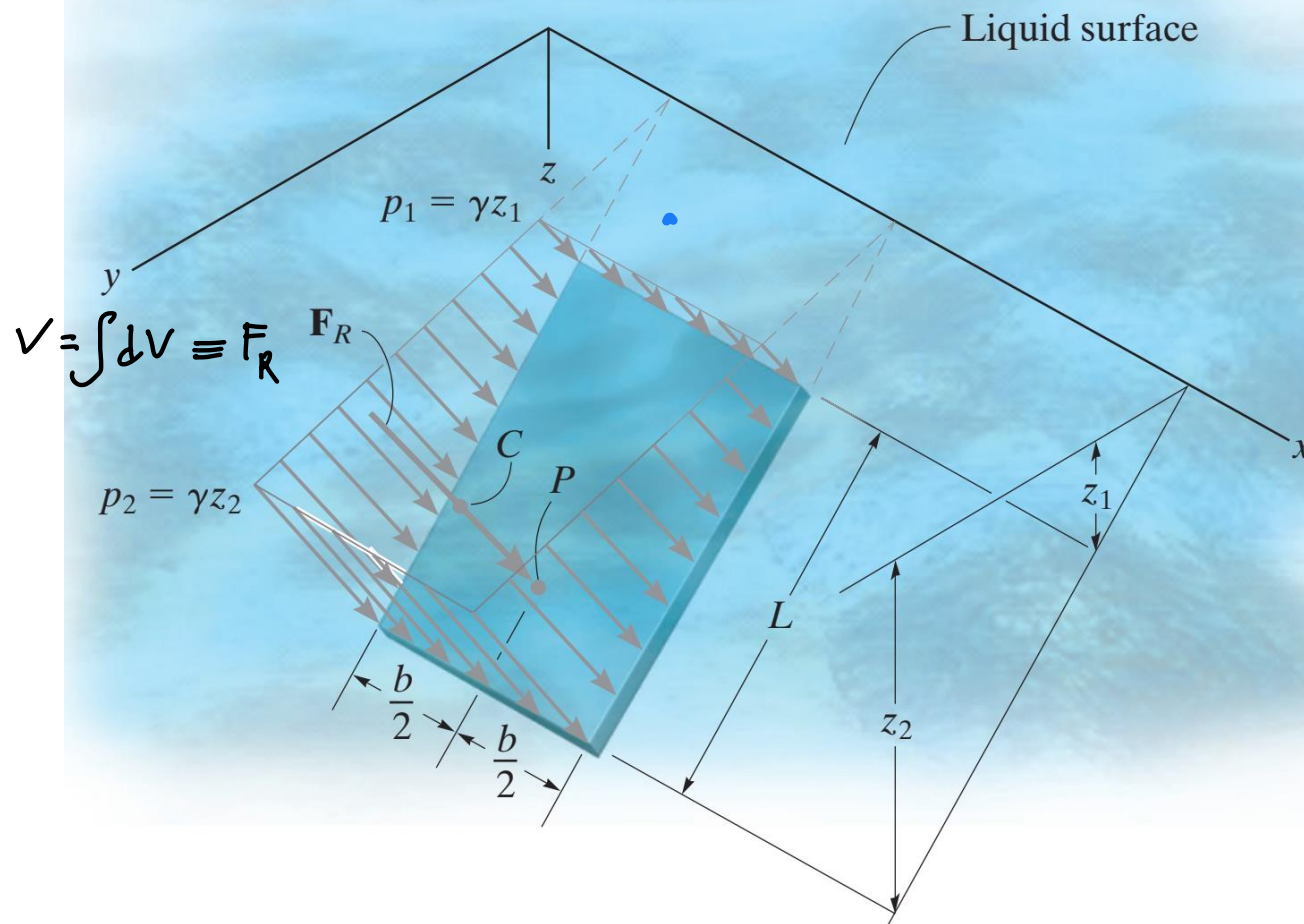
$$p = \gamma z = \rho g z$$

Fluid Pressure

The walls of the tank must be designed to support the pressure loading of the liquid that is contained within it.

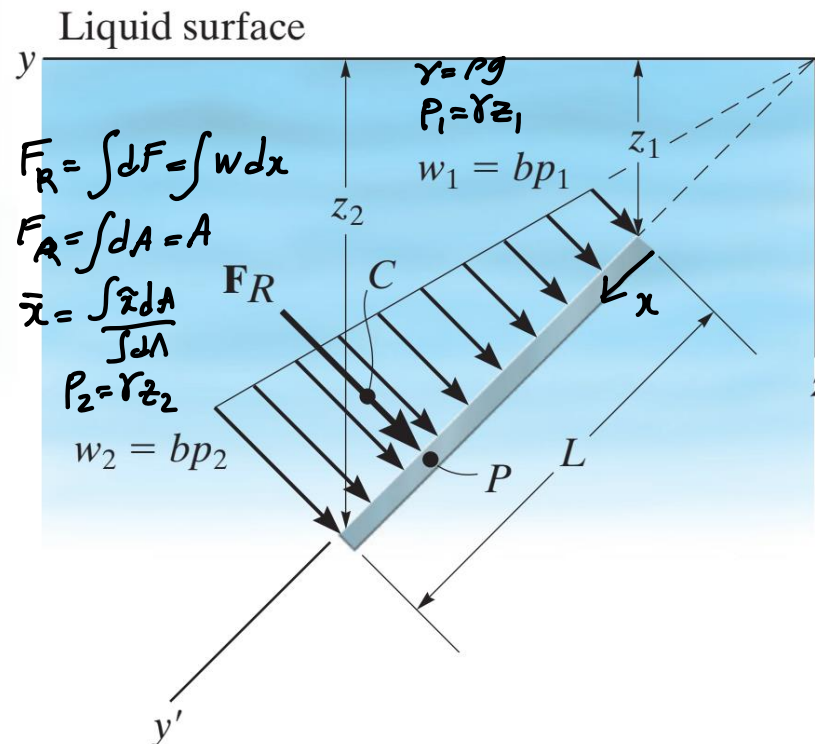


Flat Plate of Constant Width

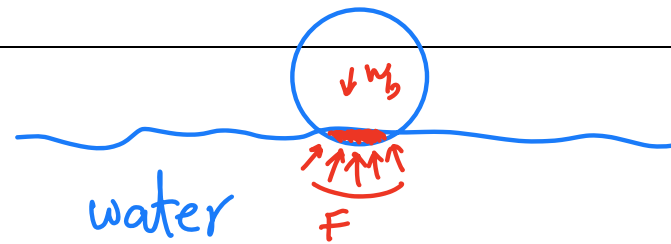


Flat Plate of Constant Width

The magnitude of the *resultant force* $\mathbf{F}_R$ is equal to the *volume* of the loading diagram, and $\mathbf{F}_R$ has a *line of action* that passes through this *volume's centroid* C . As a result, $\mathbf{F}_R$ acts on the plate at point P , called the *center of pressure*.

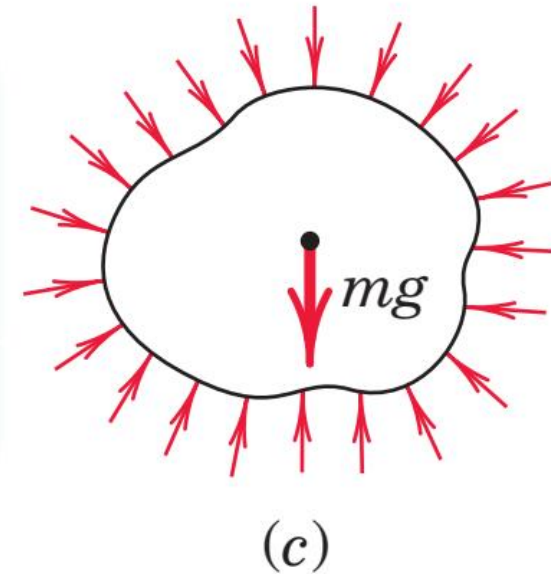
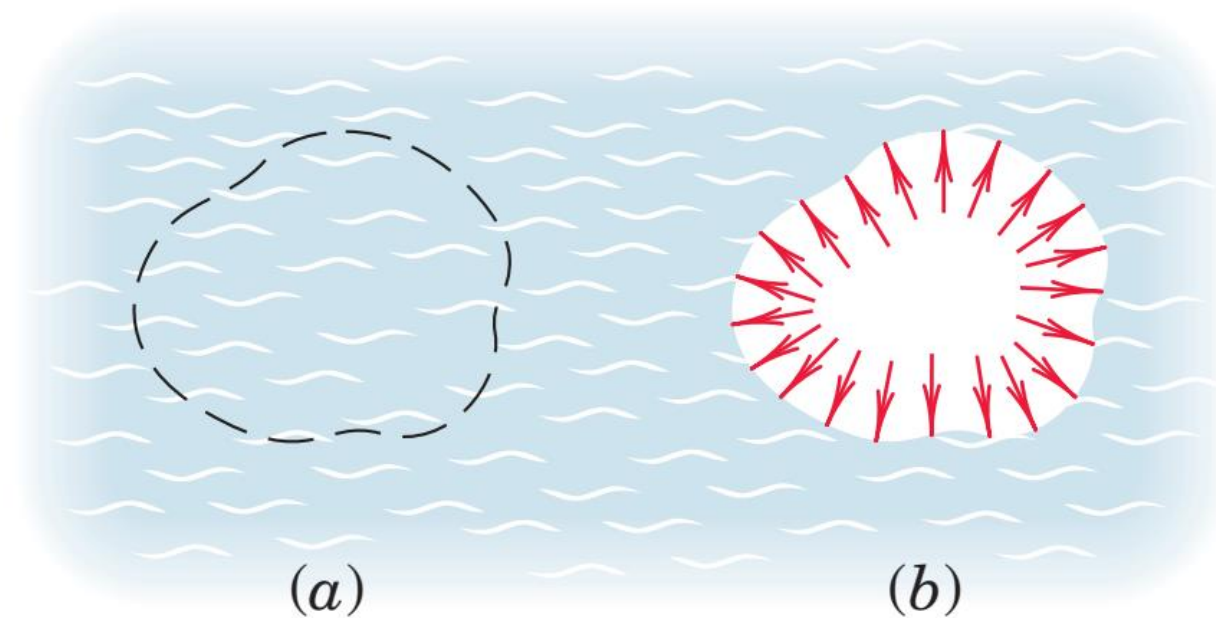


Principle of Buoyancy



$$W_b = \rho_b V_b g$$

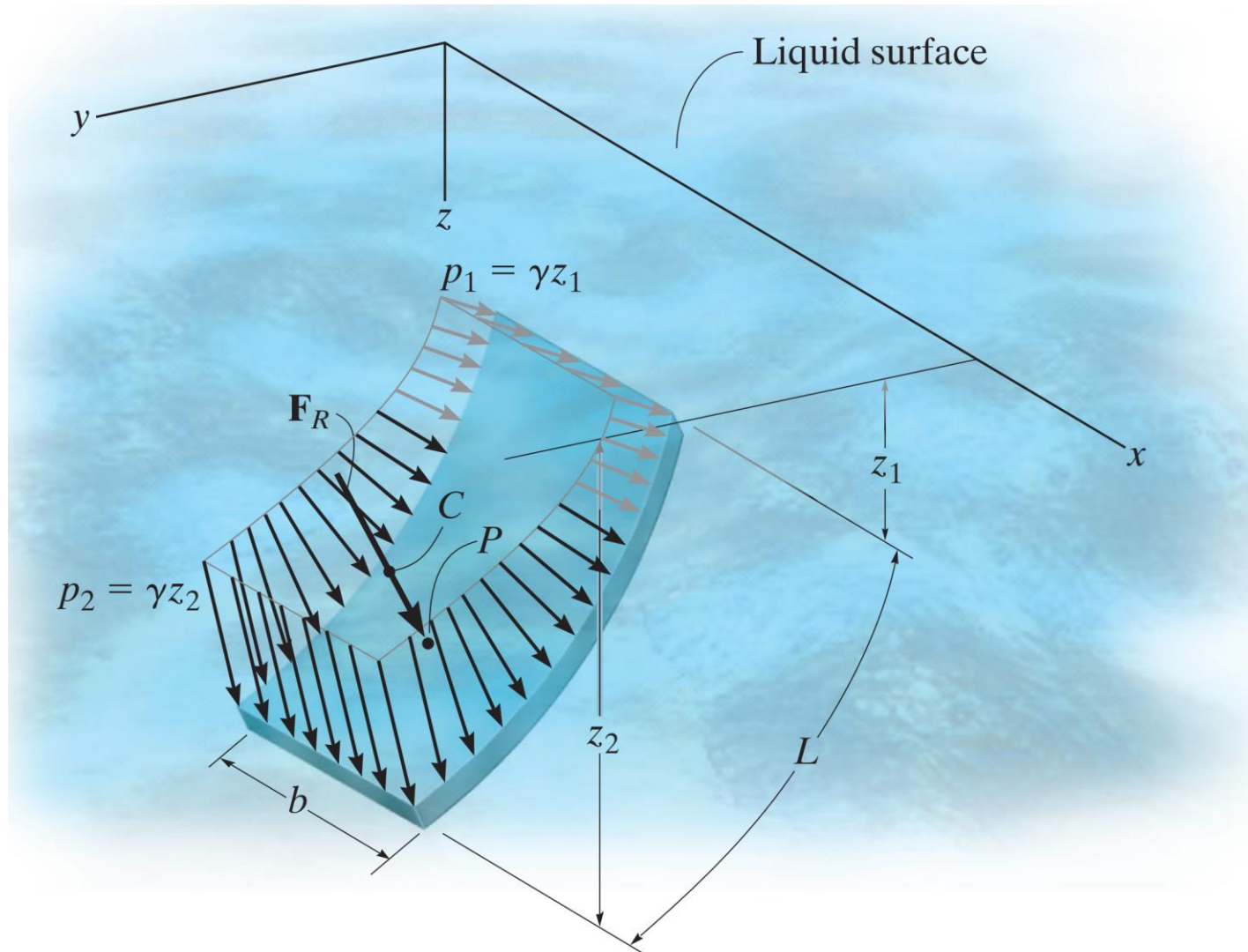
$$F = W_b / F = \rho_w g V_{\text{red part}}$$



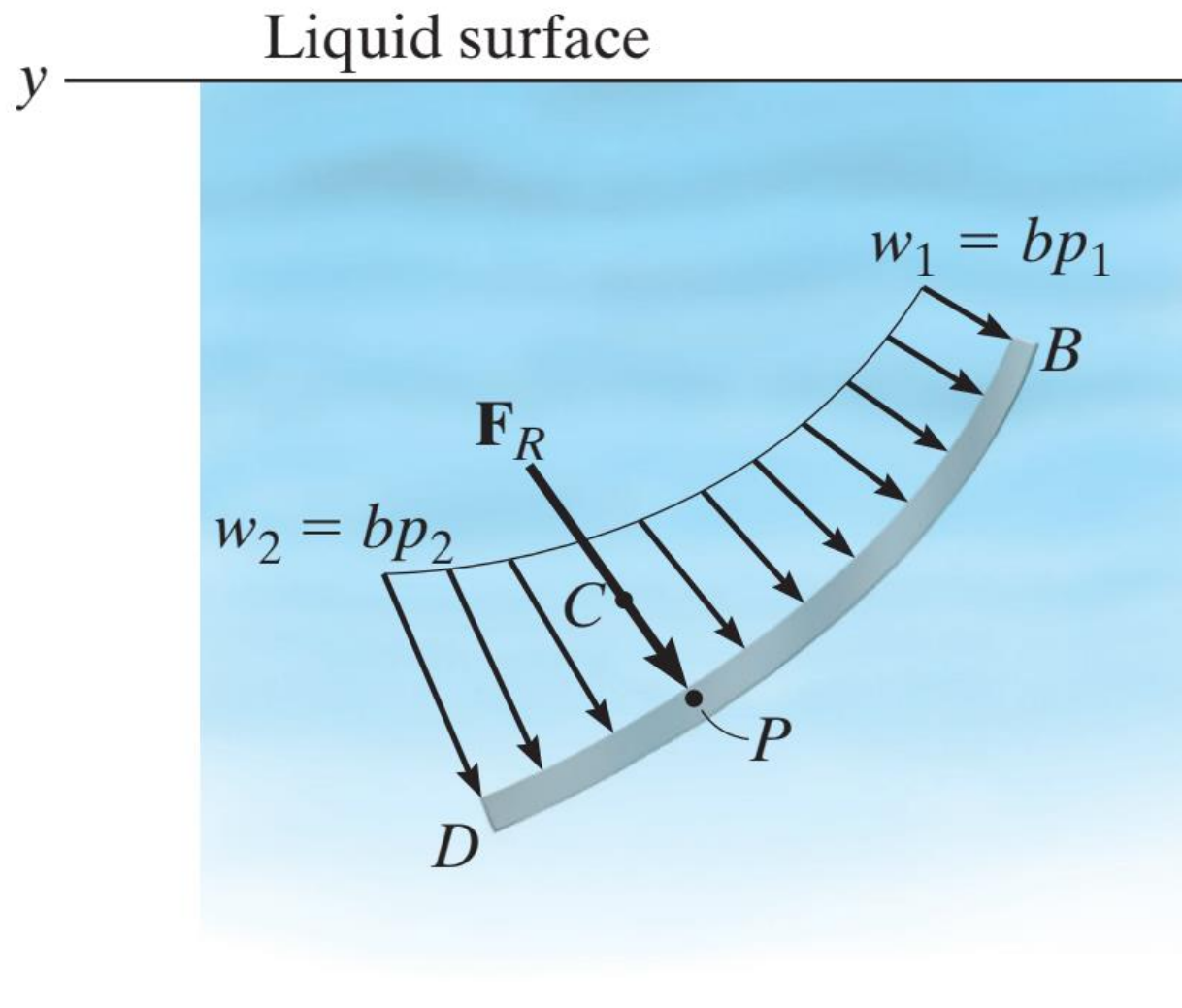
The force is called the force of *buoyancy*:

$$F = \rho g V$$

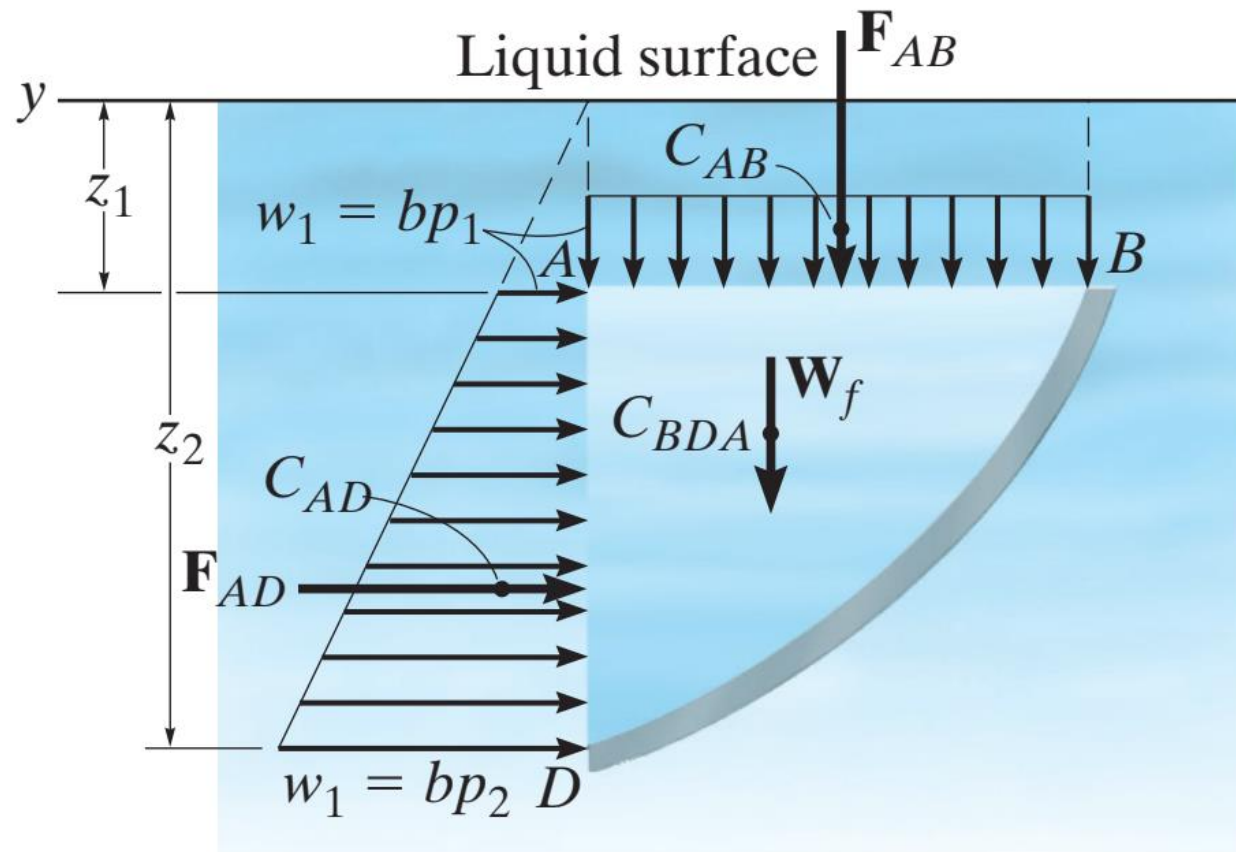
Curved Plate of Constant Width



Curved Plate of Constant Width

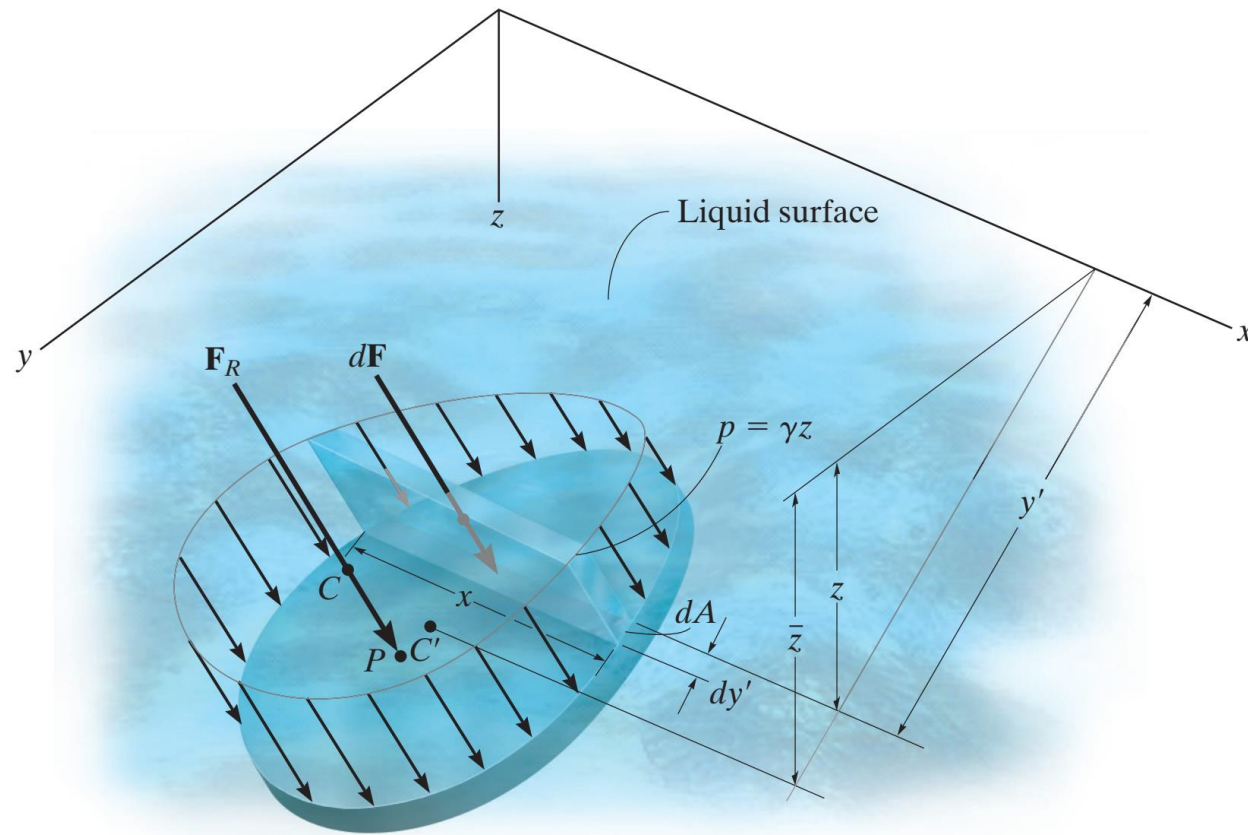


Curved Plate of Constant Width



$$\mathbf{F}_R = \Sigma \mathbf{F} = \mathbf{F}_{AD} + \mathbf{F}_{AB} + \mathbf{W}_f$$

Flat Plate of Variable Width



$$F_R = \int dF = \gamma \int z dA$$



The resultant force of the water pressure and its location on the elliptical back plate of this tank truck must be determined by integration.

Example: Determine the magnitude and location of the resultant hydrostatic force acting on the submerged rectangular plate AB shown below. The plate has a width of 1.5 m; $\rho_w = 1000 \text{ kg/m}^3$.

Diagram showing a submerged rectangular plate AB in a fluid. The plate has a width of 1.5 m and a height of 3 m. The top edge A is 2 m below the surface, and the bottom edge B is 5 m below the surface. The fluid has a density $\rho_w = 1000 \text{ kg/m}^3$ and gravity $g = 10 \text{ m/s}^2$.

Handwritten equations for the resultant force and its location:

$$F_R = A = \frac{(w_A + w_B)h}{2}$$

Where is F_R ?

$$\bar{x} = \frac{\int x dA}{\int dA}$$

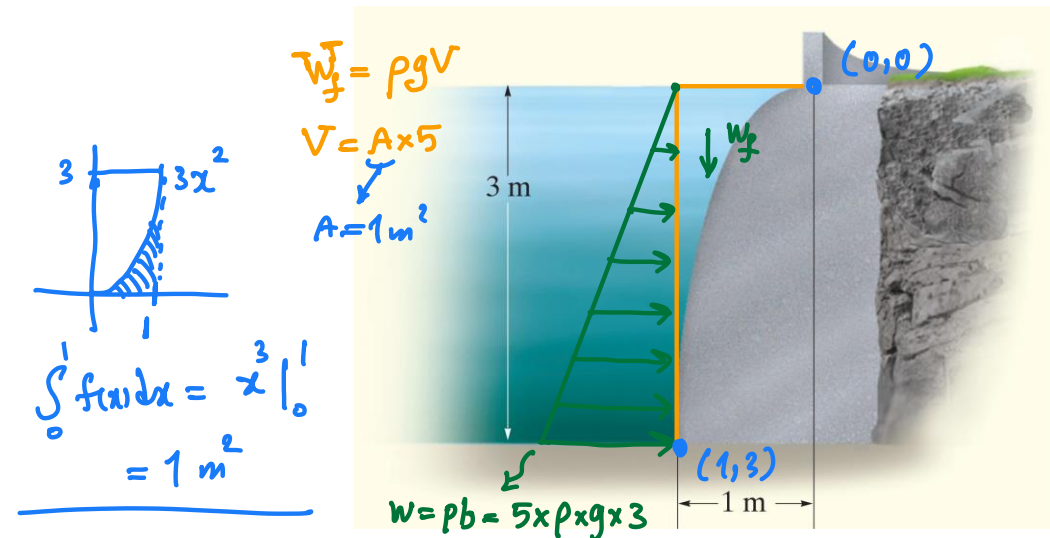
$$\bar{x} = \frac{\bar{x}_1 A_1 + \bar{x}_2 A_2}{A_1 + A_2} =$$

$$\bar{x} = \frac{1.5(w_A \times 3) + 2(w_B - w_A) \frac{3}{2}}{3w_A + (w_B - w_A) \frac{3}{2}}$$

Diagram labels and calculations:

- Top edge A : $P_A = \gamma z = 1000 \times 10 \times 2$, $w_A = b P_A$
- Bottom edge B : $w_B = b P_B$, $P_B = 1000 \times 10 \times 5$
- Height of plate: $3 \text{ m} = h$
- Width of plate: 1.5 m
- Depth of top edge: 2 m
- Depth of bottom edge: 5 m
- Gravity: $g = 10$

Example: Determine the magnitude of the resultant hydrostatic force acting on the surface of a seawall shaped in the form of a parabola, as shown below. The wall is 5 m long; $\rho_w = 1020 \text{ kg/m}^3$.



Example: Determine the magnitude and location of the resultant hydrostatic force acting on the triangular end plates of the water trough shown below; $\rho_w = 1000 \text{ kg/m}^3$.

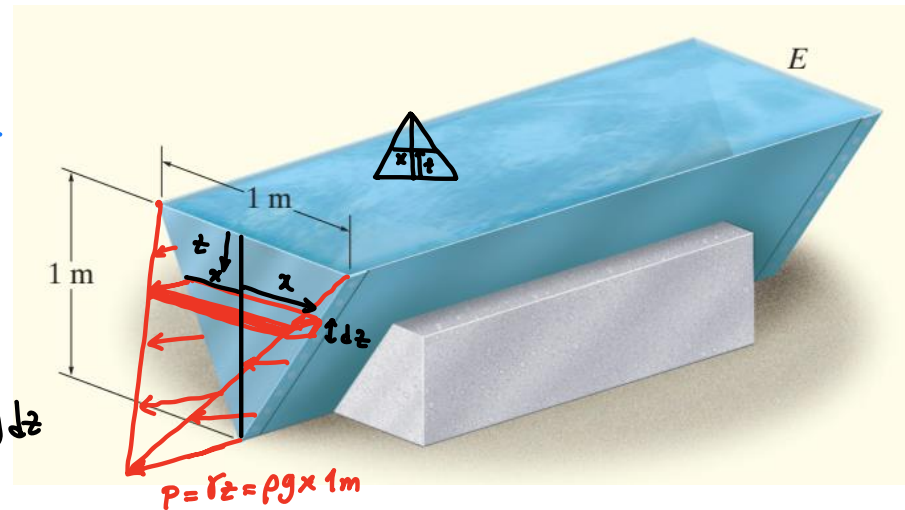
$$F_R = \int p dA = \int dV$$

$$\frac{h-z}{x} = \frac{h}{0.5}$$

$$x = 0.5 \left(1 - \frac{z}{h}\right)$$

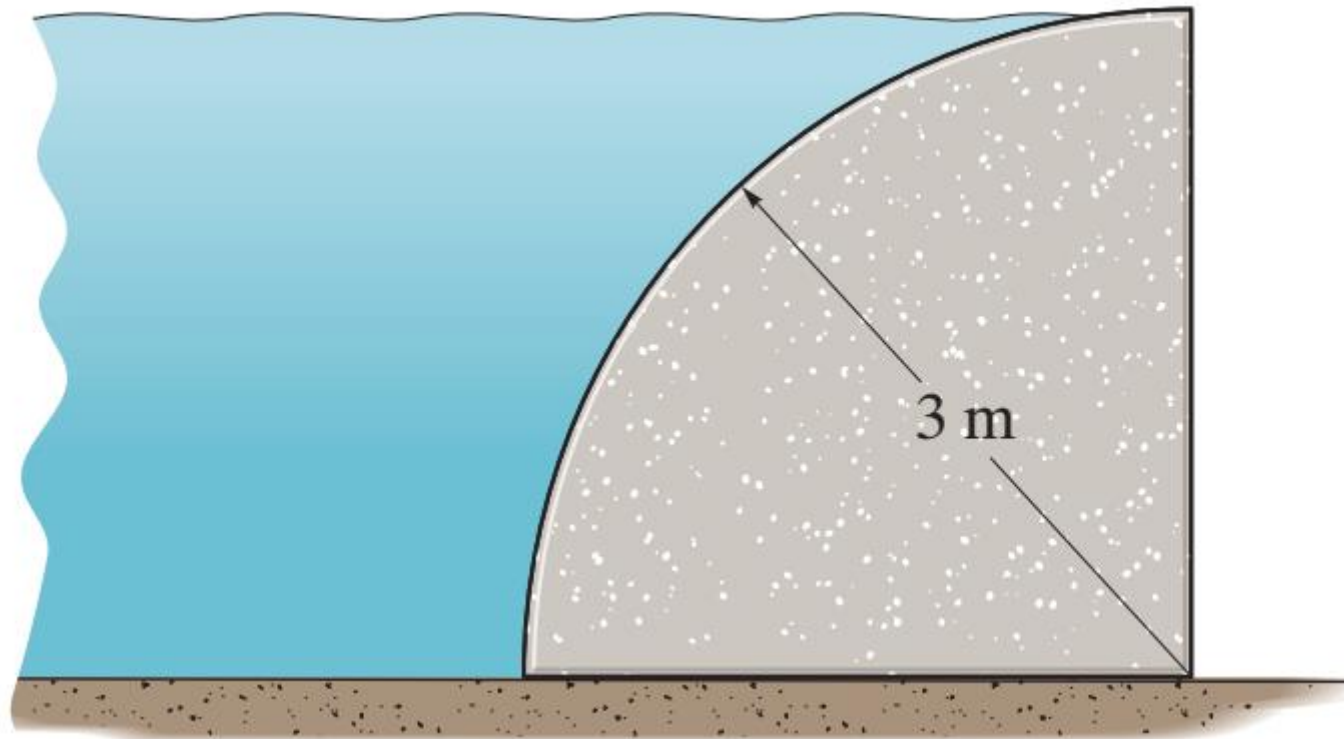
$$F_R = \int \gamma z (2x dz)$$

$$F_R = \int_0^h 2\gamma z \left(\frac{1}{2} \left(1 - \frac{z}{h}\right)\right) dz$$



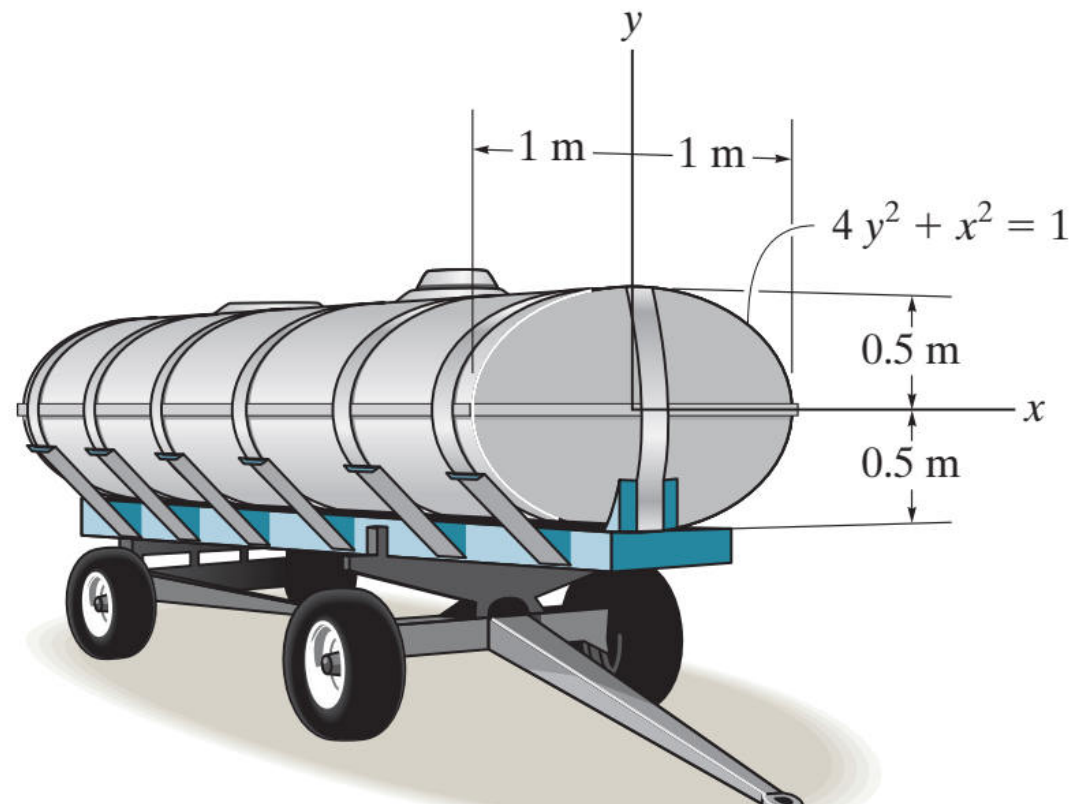
Practice at Home

1. The concrete dam in the shape of a quarter circle. Determine the magnitude of the resultant hydrostatic force that acts on the dam per meter of length. The density of water is $\rho_w = 1 \text{ Mg/m}^3$.



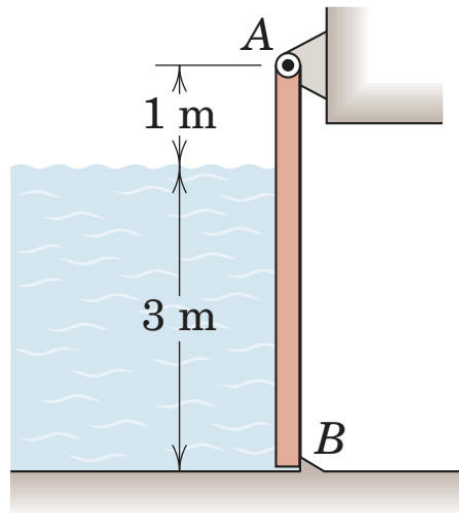
Practice at Home

2. The tank is filled with a liquid that has a density of 900 kg/m^3 . Determine the resultant force that it exerts on the elliptical end plate, and the location of the center of pressure, measured from the x axis.



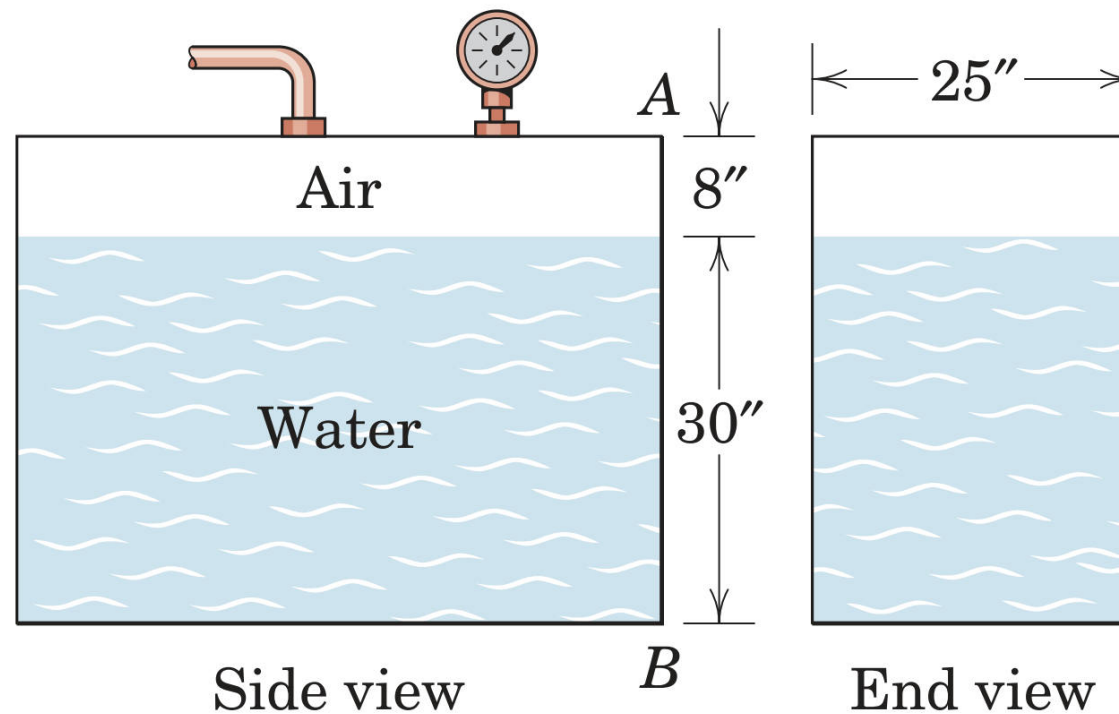
Practice at Home

3. A rectangular plate, shown in vertical section AB , is 4 m high and 6 m wide (normal to the plane of the paper) and blocks the end of a fresh-water channel 3 m deep. The plate is hinged about a horizontal axis along its upper edge through A and is restrained from opening by the fixed ridge B which bears horizontally against the lower edge of the plate. Find force B exerted on the plate by the ridge.



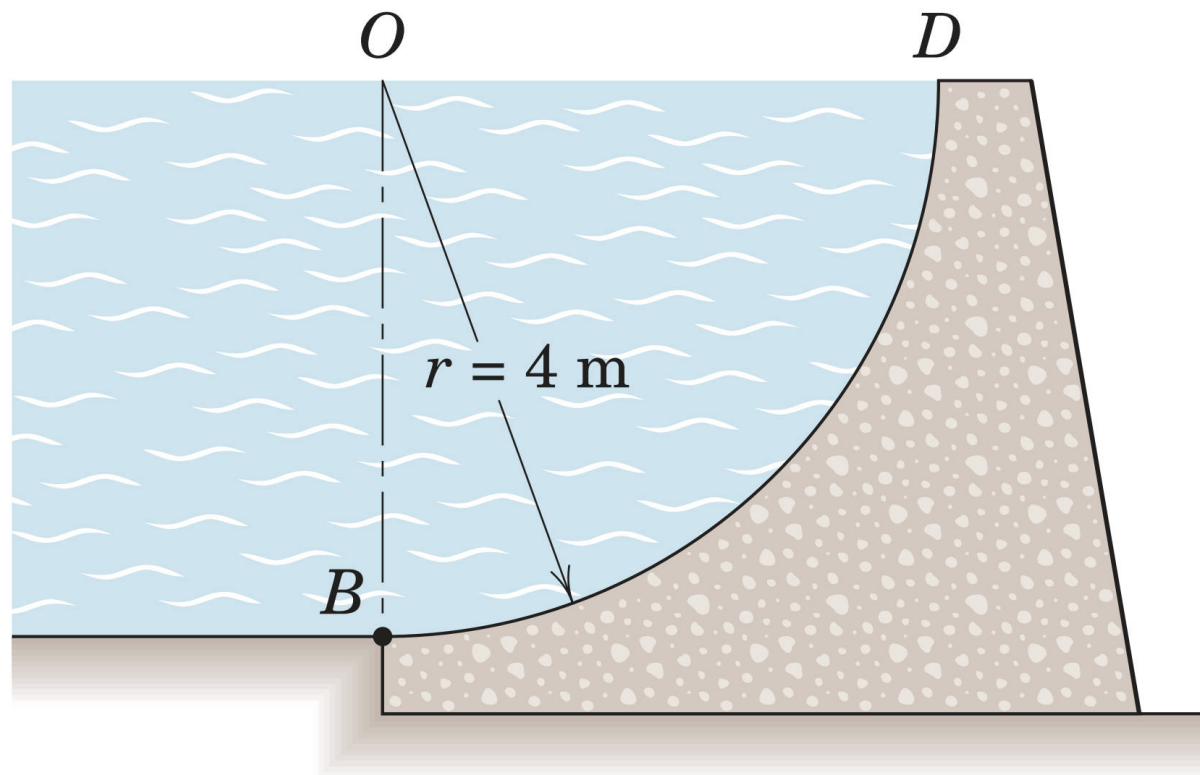
Practice at Home

4. The air space in the closed fresh-water tank is maintained at a pressure of 0.80 lb/in^2 (above atmospheric). Determine the resultant force R exerted by the air and water on the end of the tank.



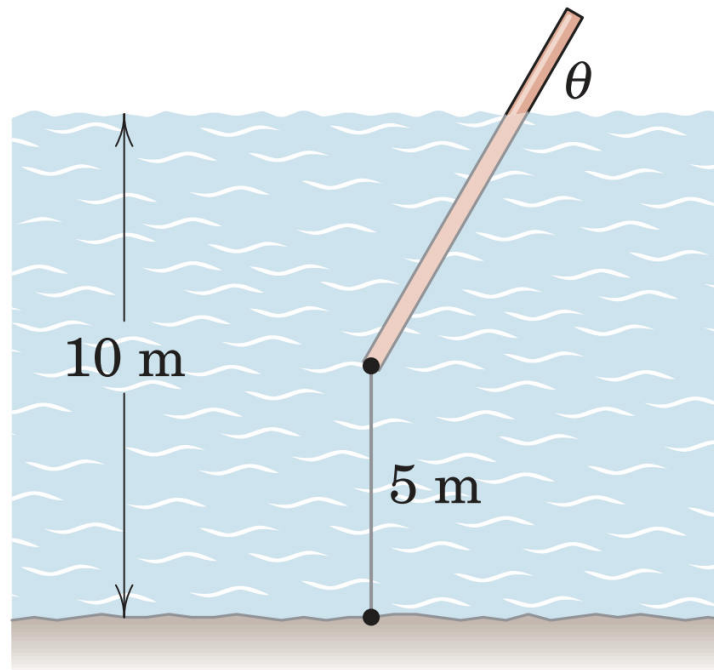
Practice at Home

5. Determine completely the resultant force R exerted on the cylindrical dam surface by the water. The density of fresh water is 1 Mg/m^3 , and the dam has a length b , normal to the paper, of 30 m.



Practice at Home

6. A buoy in the form of a uniform 8-m pole 0.2 m in diameter has a mass of 200 kg and is secured at its lower end to the bottom of a fresh-water lake with 5 m of cable. If the depth of the water is 10 m, calculate the angle made by the pole with the horizontal.



Practice at Home

7. A small remotely controlled robotic vehicle is shown below. The 200-m tether has a slight negative buoyancy in that a net downward force of 0.025 N per meter of its length acts on it. Using its onboard variable horizontal and vertical thrusters, the vehicle maintains the constant 10-m depth as it moves slowly to the right. If the maximum horizontal thrust is 10 N and the maximum vertical thrust is 7 N, determine the maximum permissible value of the distance d and state which thruster is the limiting one.

